

Manvi Vijay Gawande

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MS Data Science candidate at Indiana University Bloomington (May 2026) with hands-on experience building multi-agent AI systems, RAG pipelines, and NLP models at scale. Seeking full-time roles in Machine Learning Engineering, Applied AI, or Data Science.

EDUCATION

Indiana University, Bloomington <i>Master of Science in Data Science</i> Relevant Coursework: Applied Machine Learning, Intro to LLMs, Social Media Mining, Network Science, Data Visualization	May 2026 GPA: 3.856/4.0
Savitribai Phule Pune University <i>Bachelor of Engineering in Computer Engineering, Honors in Machine Learning and Artificial Intelligence</i> Relevant Coursework: Machine Learning, Data Science & Big Data Analytics, Cloud Computing, Database Management Systems. Awarded Excellence in Academics for top score among 160 students.	May 2024 GPA: 3.87/4.0

WORK EXPERIENCE

Applied ML Research Analyst at Indiana University Bloomington - Engineered end-to-end web scraping pipelines to collect and preprocess ~75,000 multimodal records on ivory artifacts and trade, enabling structured analysis of large-scale, heterogeneous historical and transactional datasets. - Developed recommendation systems and predictive models to identify patterns across complex ivory datasets; implemented a QWEN-powered RAG pipeline to streamline qualitative research workflows and accelerate domain-specific insight generation.	January 2026–Present
Research Assistant at Soda Lab - Analyzed 15 years of Reddit discourse (2008–2023) across 500K+ posts using LLMs and embedding models to detect belief formation patterns and behavioral shifts in online communities. - Built an NLP pipeline leveraging transformer-based embeddings to identify and predict mental-health–related signals from large-scale social media data, surfacing statistically significant linguistic markers.	September 2025–Present
Data Analyst Intern at S.K. Technovations - Performed exploratory data analysis on a dataset of over 20,000 rows, utilizing SQL for data extraction, transformation, and loading (ETL) to identify and streamline trends in customer behavior. - Automated data quality workflows, cutting processing time by 30% and saving 8 hours weekly.	November 2022–February 2023
Data Science Intern at Let's Grow More - Built a sales predictive model that increased accuracy by 13%, achieving an overall model accuracy of 91.75% - Created 15+ Tableau dashboards, delivering key insights from large-scale sales data (500K+ rows) that influenced quarterly strategy and improved revenue tracking. Used advanced Excel functions for preliminary data cleaning and trend analysis before dashboard design.	August 2022–October 2022

PROJECTS

FinGuard AI: Adaptive, Explainable Fraud Detection System <i>Technologies used: Python, LLaMA 3.1, Thompson Sampling, RAG, Async Multi-Agent Architecture</i> - Built FinGuard AI, a multi-agent fraud detection framework that performs continuous online A/B testing across 4 detection–explanation strategy pairs, using Thompson Sampling to adapt in real time and converge to dominant strategies within ~1–2k transactions without offline retraining. - Achieved 8–15% improvement in precision at fixed recall and ~20% reduction in false positives over LLM-only baselines on highly imbalanced datasets (Kaggle Credit Card Fraud, PaySim), while generating 100% auditable, regulation-grounded explanations via RAG over FinCEN and U.S. AML guidance.
Multi-Model Energy Consumption Forecasting and Benchmarking <i>Technologies used: Python, PyTorch, Scikit-learn, XGBoost, LightGBM, LSTM, Optuna, Statsmodels, Streamlit, Pandas, Matplotlib, Seaborn</i> - Engineered 50+ temporal and weather-based features for 19k+ records, driving an 18% MAPE improvement over statistical baselines. - Optimized an ML stack (XGBoost, LightGBM, LSTM) via Optuna, slashing the error rate from 153% to 22.3% by mitigating overfitting. - Developed an end-to-end pipeline and Streamlit dashboard to visualize real-time forecasts and benchmarks for a PyTorch-based LSTM.
E-Commerce Customer Segmentation and RFM Analysis <i>Technologies used: SQL (SQLite), Python(Pandas, Numpy, Plotly), A/B Testing, Tableau, Streamlit, BigQuery, GCP, dbt</i> - Built a Google BigQuery + dbt pipeline processing 100K+ records across 8 tables with 55 automated data quality tests. - Engineered an RFM model using NTILE window functions to segment 96K+ customers into 5 personas with BigQuery partitioning for query optimization. - Validated two A/B tests ($p=0.0003$, $p<0.00001$) confirming a 20% discount lifts At-Risk reactivation and \$150 shipping threshold drives ~10% AOV growth.
CineMatch : Personalized Movie Recommendation System <i>Technologies used: Python, TensorFlow/Keras, PySpark, Kafka, FastAPI, Streamlit, Docker, GitHub Actions</i> - Built a Two-Tower neural network on 100K+ ratings using PySpark-engineered features, achieving HR@10 of 0.77 and NDCG@10 of 0.49 across 9,700+ movies. - Solved cold-start for new users via genre-matched embedding averaging; pre-computed item embeddings at training time for low-latency inference. - Deployed a FastAPI + Streamlit app containerized with Docker and automated via GitHub Actions CI/CD pipeline.

TECHNICAL SKILLS

Languages: Python, SQL, R, NoSQL, Bash/Shell
ML & AI: Machine Learning, Deep Learning, NLP, RAG, Multi-Agent Systems, A/B Testing, Statistical Modeling, Embedding Models, MLOps
Frameworks & Libraries: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, HuggingFace, Sentence-Transformers, Milvus, FastAPI, Apache Spark (PySpark), Streamlit, Kafka, Optuna, Pandas/NumPy
Tools & Platforms: Tableau, Power BI, AWS, GCP, Docker, dbt, Git/GitHub, CI/CD(Github Actions), SQLite

EXTRACURRICULAR ACTIVITIES

Machine Learning Head, Google Developer Student ClubXPCCOER Organized workshops or training sessions for 200 students on machine learning for club members, fostering skill development and interest in the field.	August 2022–September 2023
3rd Place, Luddy Hackathon – Indiana University Bloomington Awarded 3rd place for developing Bibble, an AI-powered meeting companion that transforms multilingual video meetings into speaker-attributed transcripts, summaries, and emotional insights.	April 2025